Shell, Make, and CI Style Guide

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This guide covers shell scripts, Makefiles, and GitHub workflows. Agents follow `.setup/ai/style/shell.md`, which follows this guide and is changed together with it. All checks in `make lint` are warnings. `make lint` runs shellcheck when the image has it; CI always does.

1 Shell

- Start with `#!/usr/bin/env bash`, then a comment with what the script does and a usage line.
- Set `set -Eeuo pipefail` before the first command. A helper that is only sourced says "sourced" in its header and sets nothing. A command that may fail on purpose is handled where it runs.
- Quote every expansion, and use `"$@"`.
- Never hard-code a user's home or scratch folder in a template.
- Check a path before `rm -rf`: never build it from a variable that can be empty.
- Scripts are shellcheck-clean; a disabled check says why.

2 Long Runs

A run that must outlive the session starts detached, writes a status file (`running`, then `done` or `STOPPED: <reason>`) and a log next to it, refuses a second run while the status says `running`, traps

signals so that the status names the reason, and takes a `[first]` argument to resume after a failure.

3 Make

- Targets assume they already run inside the container: a recipe never calls `apptainer`.
- Each target is a short call to one R function or script.
- Use portable make: R CMD check warns about GNU extensions under `inst`.
- A target that needs SSH has a host script, since the container has no keys on Roar.

4 CI

- Use a release tag for an action, never a moving branch, and one version of each action across workflows. Use a commit SHA only for an action with no release tag.
- Jobs that use the container read it from the lock, so that a build uses the release's image.
- Tokens come only from Actions secrets, never from a file.
- `Template()` writes the workflows: change them in rProject's `inst/template`.

References

R Core Team. (2026). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing. Vienna, Austria. https://www.R-project.org/